

INHERITED TUBULAR DISORDERS

----- Active space -----

Clinical Presentation of ITD's :

↓ Reabsorption of	Presentation
1. Water	Polyuria → Compensatory polydipsia ($>2\text{L}/\text{m}^2/\text{d}$)
2. Electrolytes	Electrolyte imbalances
3. Nutrients	Failure to thrive

List of ITD's :

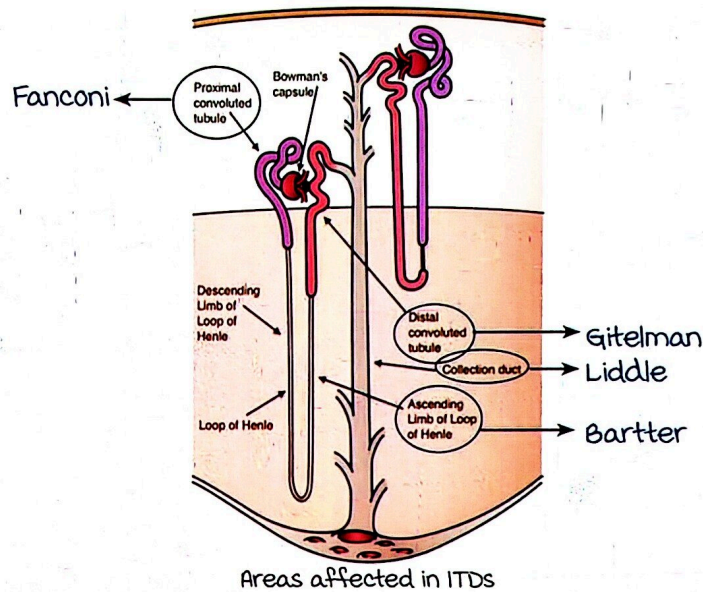
mnemonic : **FAB**ulous **G**littering Liquid.

Fanconi syndrome.

Bartter syndrome.

Gitelman syndrome.

Liddle syndrome.

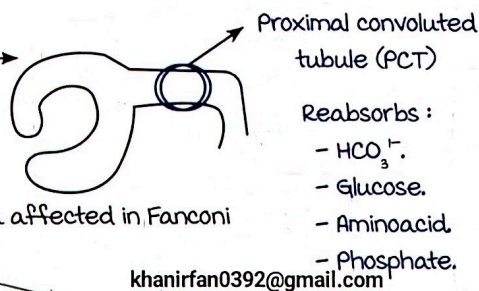


Fanconi Syndrome

00:02:13

Defect :

- Type II Renal Tubular Acidosis (RTA) (Generalized proximal tubule dysfunction)



Etiopathogenesis/clinical features :

1. Impaired reabsorption of HCO_3^-

Loss of HCO_3^- in urine

Metabolic acidosis (MA) with normal Anion gap
(w/t compensatory ↑ in Cl^-)

- Glycosuria.
- Aminoaciduria.
- Phosphaturia.

Note :

Anion gap : $\text{Cations} - \text{Anions} = \text{Na} - (\text{Cl} + \text{HCO}_3^-)$

Normal AG : $8-12 \text{ mEq/l}$

----- Active space -----

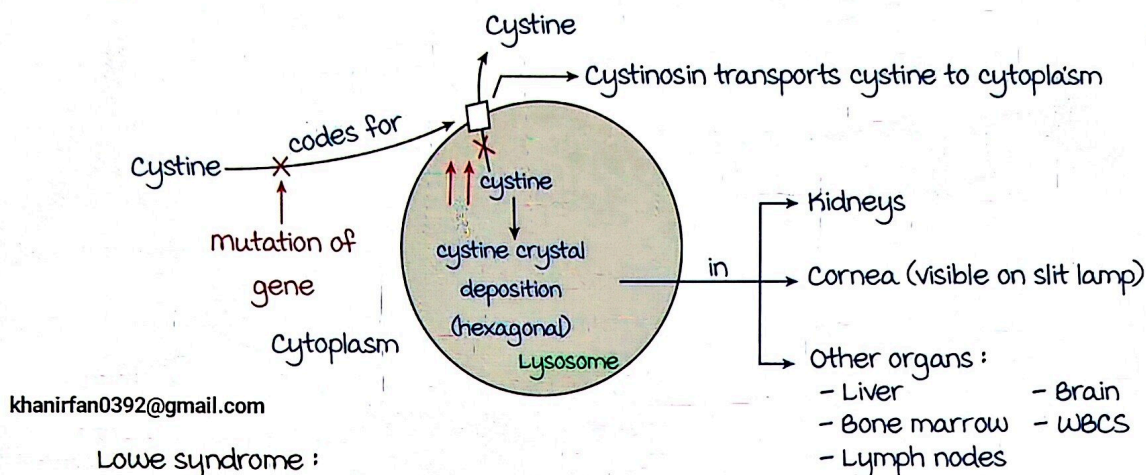
5. Urine pH :

- Acidic : <5.5 (D/t H^+ secretion & normal acidification in distal convoluted tubule).

Genetic associations :

1. Cystinosis.
2. Lowe syndrome.
3. Wilson disease.
4. Galactosemia.

Cystinosis :



Lowe syndrome :

- Gene defect : **OCRL** (Oculo-cerebro-renal Lowe) gene.
- Clinical manifestation : OCR syndrome.
 - Ocular : congenital cataract, buphthalmos.
 - Cerebral : Hypotonia, low IQ.
 - Renal : Fanconi syndrome.

Bartter Syndrome

00:12:57

- Clinical manifestations/metabolic abnormalities.
- Onset : Infancy.
- Affected area : Ascending Loop of Henle.
- Mode of inheritance : Autosomal recessive.

1. Na-K-2Cl cotransporter :

Excretion of Na^+ , K^+ , Cl^- and Ca^{2+} in urine

Depletion of fluid

Stimulation of RAAS

(↑ Na absorption)

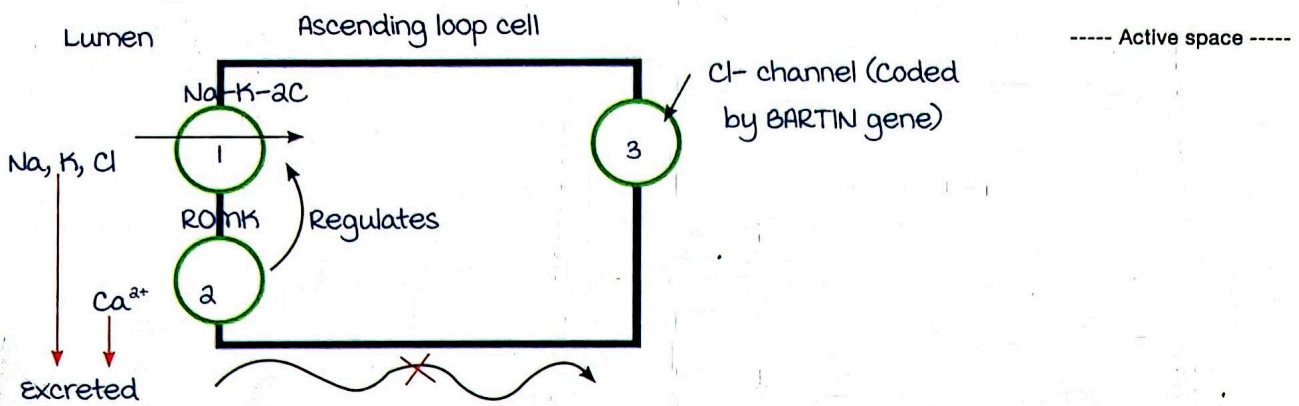
Hypercalciuria

Stone
(Nephrocalcinosis B)

RAAS : Renin-angiotensin aldosterone system

↑ K^+ excretion, H^+ excretion → Hypokalemia, hypochloremia, metabolic alkalosis2. ROMK (Renal outer medullary K^+) channel :3. Cl^- channel :

- Also expressed in cochlea.
- manifests as **SNHL** (Sensorineural hearing loss) in **BARTIN** gene defect.



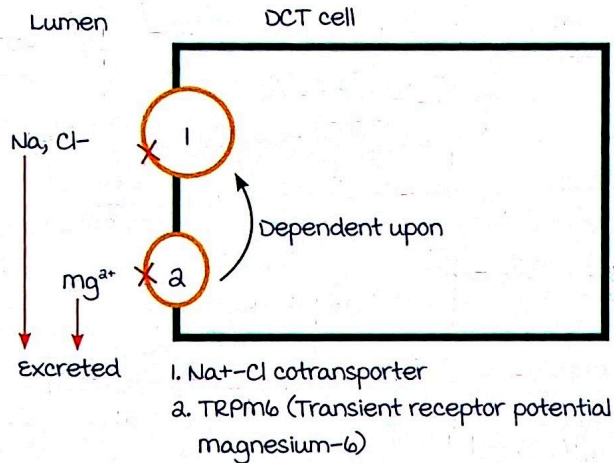
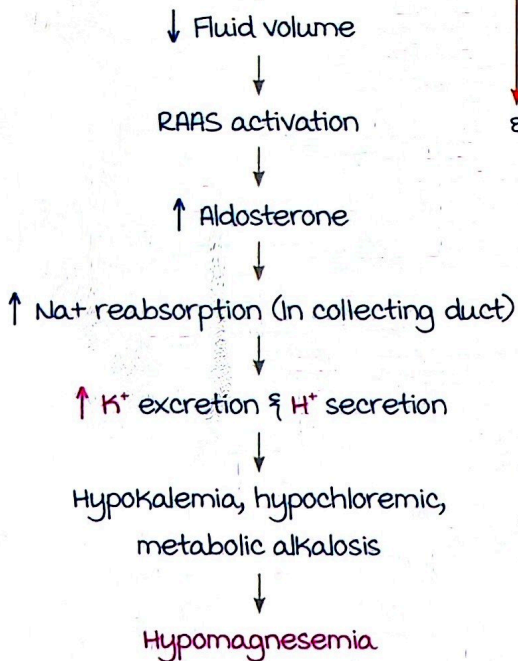
Gitelman Syndrome

00:20:23

- mode of inheritance : Autosomal Recessive.
- Affected region : **DCT**.
- Onset : Older child.
- Consequences :

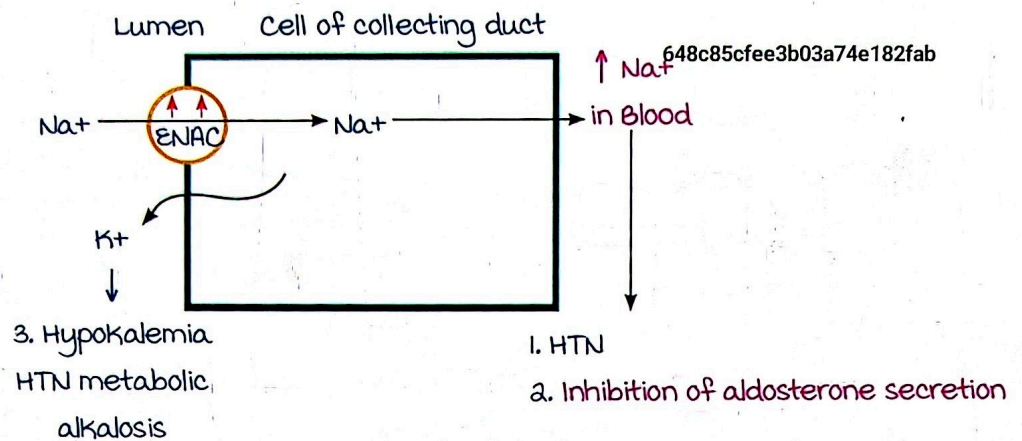
Excretion of Na^+ , Cl^- and Mg^{2+} in urine

648c85cfee3b03a74e182fab



Liddle Syndrome

- mode of inheritance : **Autosomal dominant**.
- Gene defect : $SCNN1_b$ & $SCNN1_a$ gene.
 - SCN AKA **ENAC** (Epithelial Na channel).
 - Overexpression/gain of function mutation of gene.



Evaluation of ITDs

